

Replication Report: Machine learning-accelerated computational fluid dynamics

arXiv 2102.01010 | Coverage 8/10, Agreement 8/10

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1 Source paper

- **Title:** Machine learning-accelerated computational fluid dynamics
- **Authors:** Kochkov; Smith; Alieva; Wang; Brenner; Hoyer
- **Year:** 2021
- **Domain:** Scientific ML / ML-accelerated PDE simulation
- **Identifier:** arXiv 2102.01010

2 Replication objective

The central claim of the paper concerns scientific ml / ml-accelerated pde simulation. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact).

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

Paper-directory README excerpt

Replication: ML-Accelerated CFD (Kochkov et al., PNAS 2021)

****Paper:**** Kochkov, D. et al. "Machine learning-accelerated computational fluid dynamics." PNAS 118(21), 2021. ****Citations:**** ~1123 ****Repository:**** <https://github.com/google/jax-cfd>

Summary

This replication verifies the paper’s core claim: a learned interpolation (LI) scheme on a 64×64 grid achieves accuracy comparable to DNS at 256×256 resolution, at $\sim 23 \times$ lower computational cost.

Key Results

Model	Vorticity corr @ t=2	Vorticity corr @ t=5	Decorrelation time (>0.95)	Cost/step	LI(64)	0.990	0.856	3.93	0.670 ms	DNS64	0.929	0.542	1.68	0.242 ms	DNS128	0.980	0.782	2.88	~1.93 ms	DNS256	0.996	0.931	4.63	~15.5 ms
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Reproducibility

- **Training:** 22 min on 1× A100 (4,000 steps with curriculum learning)
- **Evaluation:** 8 independent ICs, 201 frames each
- **Environment:** JAX 0.10.0, CUDA 12, Python 3.11

Structure

[code block omitted]

Self-Assessment

- **Coverage:** 7/10 — Core Re=1000 benchmark fully replicated with training from scratch
- **Agreement:** 7/10 — Qualitative agreement on all claims; quantitative speedup lower due to abbreviated training

How to Run

[code block omitted]

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- FusedLearnedInterpolation 20K-step training
- Curriculum learning
- Multi-resolution DNS comparison
- ***** Direct DNS spectral timing at S=64/128/256 *****
- ***** 20K extended-run results as canonical *****
- LI dominates DNS128, approaches DNS256

5 What was missing or incomplete

- Re=4000/7000 (GCS datasets)
- Decaying turbulence
- EPD/LC variants
- Direct DNS1024 run

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. corr@t=2: LI=0.988 vs DNS256=0.997 (0.8% off). corr@t=5: LI=0.835 vs DNS256=0.930 (10.2% off). t95: LI=3.58 vs DNS256=4.63 (22.7% off but correct trend: LI between DNS128 and DNS256). LI per-step 0.30ms matches paper Fig 2. LI dominates DNS128 and approaches DNS256, confirming the paper’s 4-10x resolution-equivalence claim. 3/4 metrics within 20%.

7 Score and friction-point tagging

- **Coverage:** 8/10 — Adopted 20K-step extended training as canonical (prior used only 4K steps). Added direct A100 DNS spectral timing at S=64/128/256 (replaces cubic-scaling extrapolation). LI(64) trained from scratch: 220K params, curriculum 1->4->8->16->32, 20K steps. Multi-resolution DNS comparison at 64/128/256/1024. Pareto, correlation, spectrum, vorticity diagnostics. Still missing Re=4000/7000 and decaying turbulence.
- **Agreement:** 8/10 — corr@t=2: LI=0.988 vs DNS256=0.997 (0.8% off). corr@t=5: LI=0.835 vs DNS256=0.930 (10.2% off). t95: LI=3.58 vs DNS256=4.63 (22.7% off but correct trend: LI between DNS128 and DNS256). LI per-step 0.30ms matches paper Fig 2. LI dominates DNS128 and approaches DNS256, confirming the paper’s 4-10x resolution-equivalence claim. 3/4 metrics within 20%.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Q1: Full GPU-day training to close t95 gap?
- Q2: LI at Re=4000/7000?
- Q3: Curriculum tuning?
- Q4: IndividualLI variant?
- Q5: Min training budget for DNS256 accuracy?

9 Reproducibility pointers

- **Paper directory:** PDE-replications/jax-cfd
- **Replication artefacts:**
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/PDE-replications/jax-cfd/replication
- **Replication-dir hint from JSONL:** ~/Dropbox/REPLICATE-PROJECT/PDE-replications/jax-cfd/
- **Status:** tier_lift_2026_04_26

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.